

IZZUL QUDSI

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PROFILE

Geoscientist with 9+ years of experience in subsurface analysis, petrophysics, and exploration workflows. Experienced in integrating geological interpretation, artificial intelligence, and digital technologies to improve subsurface understanding and exploration decision-making. Currently completing a Ph.D. in Geology focused on digital subsurface characterization and geothermal exploration. Strong background in consultancy, R&D collaboration, and transform research workflows into practical, reproducible tools for geoscientists.

Core competencies: Subsurface interpretation • Subsurface data management • Python automation • AI for geoscience • Spectral Geology • Web-apps for geoscience

EXPERIENCE

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| 2025 | Student Co-op Training, National Petroleum Technology Company (NESR) <ul style="list-style-type: none">• Supported R&D projects as data geoscientist.• Contributed to the development of a prototype software platform for core data analysis. |
| 2021-Present | Group Leader, Geovarthas.id (non-commercial group discussion) <ul style="list-style-type: none">• Led and organized the research, project and communication within the activities.• Develop web-based geoscience tools to support geoscientific analysis: Wellcast Geothermie Spectral-Box Many more http://www.geovarthas.id |
| 2019 - 2023 | Technical Geoscience Specialist, Horizon Perdana <ul style="list-style-type: none">• Delivered consultancy support for well & seismic interpretation and petrophysics.• Led the Data Management team to the initiation of the creation of the web application and its database infrastructure.• Provided technical support for Elsiwave, OpendTect and Completion Manager. |
| 2014 – 2017 | Geoscientist, Trisakti University Oil and Gas Research Group (PUKESMIGAS)
Geological mapping, core analysis, well/seismic interpretation and petrophysics. |
| 2013 – 2014 | MWD Engineer, Cougar Drilling Solution
Performed MWD surveys and supported directional drilling operations. |

TECHNICAL SKILLS

Programming/Data: Python (AI, ETL/QC, web backend), SQL, PPDM, APIs

Geoscience: Petrophysics, well log & core analysis, XRD/XRF, Hyperspectral, Seismic interpretation

Software: Petrel, Techlog, ArcGIS, ENVI, Oasis montaj, HampsonRussell, OpendTect

EDUCATION

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

2023-now: Ph.D., *Geology* | GPA: 3.83/4 (ongoing)

- Winner of GEO4.0 Digitalization in Geoscience Symposium Hackathon
- Runner-Up at the Agentic AI Hackathon 2025 by Ulster University, UK
- Top 10 winners at Dhahran Techno Valley Deep-Tech Ventures Program

University of Twente, Enschede, The Netherlands

March 2019: Master of Science, *Applied Earth Science* | GPA: 7.9/10

- Faculty of ITC Excellence Scholarship awardee

Universitas Padjadjaran, Bandung, Indonesia

February 2013: Bachelor of Engineering, *Geology* | GPA: 3.34/4

PUBLICATIONS

- 2025 Reconstructing regional subsurface thermal structure of the Okavango rift zone, northern Botswana, and its low enthalpy geothermal potential. *Geothermics*. <https://doi.org/10.1016/j.geothermics.2025.103519>
- 2025 Leveraging Generative AI and Hyperspectral Drill-Core Imaging for Fully Automated Mineral Mapping of Unconventional Reservoir. *Applied Computing and Geosciences* <https://doi.org/10.1016/j.acags.2025.100298>
- 2025 Core Digitization: A Journey to Digital Transformation of Unconventional Mudstone Reservoir. SPE Conference at Oman Petroleum & Energy Show. <http://dx.doi.org/10.2118/224825-MS>
- 2022 Mineral Mapping on Hyperspectral Imageries Using Cohesion-based Self Merging Algorithm. *Jurnal Elektronika dan Telekomunikasi* <http://dx.doi.org/10.55981/jet.507>
- 2022 Geothermal Surface Manifestation Identification Using Airborne Hyperspectral Imagery. Case Study: Davis-Schrimp Geothermal Field, Salton Sea California at Indonesian Journal on Geoscience [DOI: 10.17014/ijog.9.1.119-130](https://doi.org/10.17014/ijog.9.1.119-130)
- 2021 The Application of Parametric and Non-parametric Regression to Predict the Missing Well Log Data at Indonesian Journal on Geoscience [DOI: 10.17014/ijog.8.3.385-399](https://doi.org/10.17014/ijog.8.3.385-399)

CONFERENCES

- 2025 Fast and Robust In-Situ Quantitative Mineral Analysis of Carbonate Rocks with AI-Driven Hyperspectral Unmixing Approach. MEOS-GEO 2025.
- 2024 Deep Learning-Assisted Quantitative Mineral Mapping of The Devonian Shale Core Using Hyperspectral Imaging. GEO4.0 Digitalization in Geoscience Symposium
- 2024 Estimating the Regional Thermal Structure of Crustal Thinning Zone in Northern Botswana at Topical Geothermal Workshop by SPWLA Saudi Arabia Chapter
- 2022 Mineral Identification on Hyperspectral Imagery of Rock Samples Using Machine Learning at 1st International Conference on Upstream Energy Technology and Digitalization – Universitas Pertamina, Jakarta. <https://doi.org/10.1063/5.0208704>
- 2021 Curie Point Depth Mapping using PyCurious from Aeromagnetic Data at GeoPython Conference 2021